Tutorial 2: First-Order ODEs (Part II)

[Attempt all questions before tutorial session]

A) Bernoulli's Equation

Solve

$$2x(\ln x)y' - y = -9x^3y^3 \ln x.$$

Solution: This equation is nonlinear since it involves the term y^3. We rewrite it as the standard form

$$y' - \frac{1}{2x(\ln x)}y = -\frac{9}{2x^2}y^3,$$

which is a Bernoulli equation of order $n = 3$. Diving by y^3 yields the equivalent equation:

$$y^{-3}\frac{dy}{dx} - \frac{1}{2x(\ln x)}y^{-2} = -\frac{9}{2}x^2. \tag{1}$$

We next make the change of variables

$$u = y^{-2} \implies \frac{du}{dx} = -2y^{-3}\frac{dy}{dx} \implies y^{-3}\frac{dy}{dx} = -\frac{1}{2}\frac{du}{dx}.$$

Substituting into (1) gives

$$-\frac{1}{2}\frac{du}{dx} - \frac{1}{2x(\ln x)}u = -\frac{9}{2}x^2.$$

Rewriting it as the standard form gives

$$\frac{du}{dx} + \frac{1}{x(\ln x)}u = 9x^2,$$

which is a linear equation. The solution is

$$\begin{aligned} u(x) &= e^{-\int \frac{1}{x(\ln x)}dx}\left[\int 9x^2 e^{\int \frac{1}{x(\ln x)}dx}dx + C\right] \\ &= e^{-\ln(\ln x)}\left[\int 9x^2 e^{\ln(\ln x)}dx + C\right] \\ &= \frac{1}{\ln x}\left[9\int x^2 \ln x dx + C\right] = \frac{1}{\ln x}\left[3x^3 \ln x - x^3 + C\right]. \end{aligned}$$

Thus, the general solution is

$$y^{-2} = \frac{1}{\ln x}\Big[3x^3 \ln x - x^3 + C\Big] \iff y^2 = \frac{\ln x}{x^3(3\ln x - 1) + C}.$$

B) **Exact DE**

Determine if the following equations are exact. If so, find its general solution

a) $\left(3x^2y - 2y^2\right)dx + \left(x^3 - 4xy + 6y^2\right)dy = 0,$

b) $(\sin(xy) + xy\cos(xy) + 2x)dx + (x^2\cos(xy) + 2y)dy = 0,$

c) $x^2ydx - (xy^2 + y^3)dy = 0.$

Solution: (a) The equation is already in differential form with

$$M(x,y) = 3x^2y - 2y^2, \quad N(x,y) = x^3 - 4xy + 6y^2.$$

We test if it is exact by verifying

$$\frac{\partial M}{\partial y} = 3x^2 - 4y = \frac{\partial N}{\partial x}.$$

Therefore, this DE is exact! Then we look for a potential function u satisfying

$$\begin{cases} \dfrac{\partial u}{\partial x} = M = 3x^2y - 2y^2, \\ \dfrac{\partial u}{\partial y} = N = x^3 - 4xy + 6y^2. \end{cases}$$

From the first equation, integrating with respect to x keeping y constant, we have

$$u(x,y) = x^3y - 2xy^2 + g(y)$$

where $g(y)$ is the "constant" of integration. Substituting this into the second equation yields

$$x^3 - 4xy + g'(y) = x^3 - 4xy + 6y^2,$$

from which, we find $g'(y) = 6y^2$, i.e., $g(y) = 2y^3 + c$. We can take $c = 0$, and obtain the potential function: $u = x^3y - 2xy^2 + 2y^3$. Then the general solution is

$$x^3y - 2xy^2 + 2y^3 = C.$$

Alternative solution by inspection:

$$\begin{aligned}\left(3x^2y-2y^2\right)dx+\left(x^3-4xy+6y^2\right)dy &= \left(3x^2ydx+x^3dy\right)-\left(2y^2dx+4xydy\right)+6y^2dy\\ &= d\left(x^3y\right)-d\left(2xy^2\right)+d\left(2y^3\right)=d\left(x^3y-2xy^2+2y^3\right)\\ &= d\left(x^3y-2xy^2+2y^3+C\right)\end{aligned}$$

Then the required function is $x^3y-2xy^2+2y^3+c$. This method, called the grouping method, is based on one's ability to recognize exact differential combinations.

b) We have

$$\begin{aligned} M(x,y)&=\sin(xy)+xy\cos(xy)+2x \quad\Longrightarrow\quad \frac{\partial M}{\partial y}=2x\cos(xy)-x^2y\sin(xy),\\ N(x,y)&=x^2\cos(xy)+2y \quad\Longrightarrow\quad \frac{\partial N}{\partial x}=2x\cos(xy)-x^2y\sin(xy)=\frac{\partial M}{\partial y},\end{aligned}$$

and so the equation is exact. Hence, there exists a potential function $u(x,y)$ such that

$$\begin{aligned}\frac{\partial u}{\partial x}&=\sin(xy)+xy\cos(xy)+2x,\\ \frac{\partial u}{\partial y}&=x^2\cos(xy)+2y.\end{aligned}\tag{2}$$

In this case, the second equation is the simpler equation, and so we integrate it with respect to y, holding x fixed, to obtain

$$u(x,y)=x\sin(xy)+y^2+h(x),$$

where $h(x)$ is an arbitrary function of x. We now determine $h(x)$. Differentiating the resulting $u(x,y)$ partially with respect to x yields

$$\frac{\partial u}{\partial x}=\sin(xy)+xy\cos(xy)+h'(x).$$

Hence, we have

$$h'(x)=2x \quad\Rightarrow\quad h(x)=x^2.$$

Here, we set the integration constant to zero, since we only need one potential function

$$u(x,y)=x\sin(xy)+x^2+y^2.$$

The original equation can be written as

$$d(x\sin(xy) + x^2 + y^2) = 0,$$

and hence the general solution is

$$x\sin(xy) + x^2 + y^2 = C.$$

(c) We have that

$$M(x,y) = x^2y \implies \frac{\partial M}{\partial y} = x^2,$$

while

$$N(x,y) = -(xy^2 + y^3) \implies \frac{\partial N}{\partial x} = -y^2.$$

Since $M_y \neq N_x$, the equation is non-exact.

C) Determine an integrating factor for the equation

$$ydx - (2x + y^4)dy = 0,$$

and hence find the general solution.

Solution: Assume that the function $I(x,y)$ is an integrating factor of this equation. Multiplying the equation by $I(x,y)$ yields

$$\underbrace{Iy}_{P(x,y)}dx + \underbrace{(-(2x+y^4))I}_{Q(x,y)}dy = 0. \tag{3}$$

It is exact if and only if

$$P_y = yI_y + I = Q_x = -2I - (2x + y^4)I_x.$$

Observe that if I is only a function of y, i.e., $I_x = 0$, then we can solve

$$yI_y = -3I \implies I(y) = \frac{1}{y^3}.$$

Plugging it into the equation (3) gives

$$\frac{1}{y^2}dx - \frac{2x + y^4}{y^3}dy = 0,$$

which is exact. We now solve the potential function

$$\frac{\partial \phi}{\partial x} = \frac{1}{y^2}, \qquad \frac{\partial \phi}{\partial y} = -\frac{2x + y^4}{y^3}.$$

From the first equation, we get that

$$\phi(x, y) = \frac{x}{y^2} + h(y) \implies \frac{\partial \phi}{\partial y} = -\frac{2x}{y^3} + h'(y).$$

By the second equation,

$$h'(y) = -y \implies h(y) = -\frac{y^2}{2}.$$

Accordingly, the general solution is

$$\frac{x}{y^2} - \frac{y^2}{2} = c \implies 2x - y^4 = Cy^2.$$

D) Determine whether the given function $I(x, y) = \cos(xy)$, is an integrating factor for the DE

$$\big[\tan(xy) + xy\big]dx + x^2 dy = 0.$$

If so, find its general solution.

Solution: Multiplying the equation by $I(x, y)$ gives

$$\underbrace{\big[\sin(xy) + xy\cos(xy)\big]}_{P(x,y)}dx + \underbrace{x^2\cos(xy)}_{Q(x,y)}dy = 0.$$

Check that

$$P_y = 2x\cos(xy) - x^2 y\sin(xy) = Q_x.$$

Therefore, $\cos(xy)$ is an integrating factor of the given equation.

Note that by inspection, we find

$$d(x\sin(xy)) = \big(\sin(xy) + xy\cos(xy)\big)dx + x^2\cos(xy)dy,$$

so the general solution is $x\sin(xy) = C$.

E) **Homogeneous DE**

Solve

$$y' - x^{-1}y = x^{-1}\sqrt{x^2 - y^2}, \quad x > 0.$$

Solution: By inspection, we know that this equation could not be separable, linear or Bernoulli. We try to check if it is homogenous, and hence rewrite it as

$$\frac{dy}{dx} = \frac{y}{x} + \frac{\sqrt{x^2 - y^2}}{x} = \frac{y}{x} + \sqrt{1 - \left(\frac{y}{x}\right)^2}.$$

We recognize it as a homogenous DE, since the RHS only depends on y/x. Change the variables

$$y = xV(x) \implies \frac{dy}{dx} = x\frac{dV}{dx} + V,$$

and substitute them into the equation to eliminate y, y' :

$$x\frac{dV}{dx} + V = V + \sqrt{1 - V^2} \implies \frac{dV}{\sqrt{1 - V^2}} = \frac{dx}{x}.$$

Direct integration leads to

$$\int \frac{dV}{\sqrt{1 - V^2}} = \int \frac{dx}{x} + C \implies \sin^{-1} V = C + \ln|x|.$$

Hence, the general solution is

$$y = xV = x\sin(C + \ln x), \quad x > 0.$$

F) Solve the following equations by (i) Bernoulli's technique or (ii) finding a suitable integrating factor and then converting it to an exact equation:

$$\frac{dy}{dx} = -\frac{8x^5 + 3y^4}{4xy^3}.$$

Solution: We first rewrite it as

$$\frac{dy}{dx} + \frac{3}{4x}y = -2x^4y^{-3},$$

which we recognize as a Bernoulli equation of order $n = -3$. We can solve it by using the corresponding solution formula.

Here, we solve it by using the second method. We check for exactness and rewrite the given DE in a differential form

$$(8x^5 + 3y^4)dx + 4xy^3dy = 0.$$

We have

$$P_y = 12y^3, \quad Q_x = 4y^3,$$

so that the equation is not exact. However, we see that

$$\frac{P_y - Q_x}{Q} = 2x^{-1}.$$

Therefore, $I(x) = x^2$ is an integrating factor, which is obtained by solving

$$\frac{dI}{dx} = \frac{P_y - Q_x}{Q} I = \frac{2I}{x}.$$

Multiplying both sides by x^2 yields

$$x^2(8x^5 + 3y^4)dx + 4x^3y^3dy = 0.$$

From inspection, we find

$$x^2(8x^5 + 3y^4)dx + 4x^3y^3dy = d(x^8 + x^3y^4),$$

so the general solution is

$$x^8 + x^3y^4 = C.$$